Section Overview

- Throughput, Arithmetic Intensity, and the Roofline Model

Driving Question

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Peak throughput	35.6	$\mathrm{TFLOP/s}$
DRAM bandwidth	935.8	$\mathrm{GB/s}$

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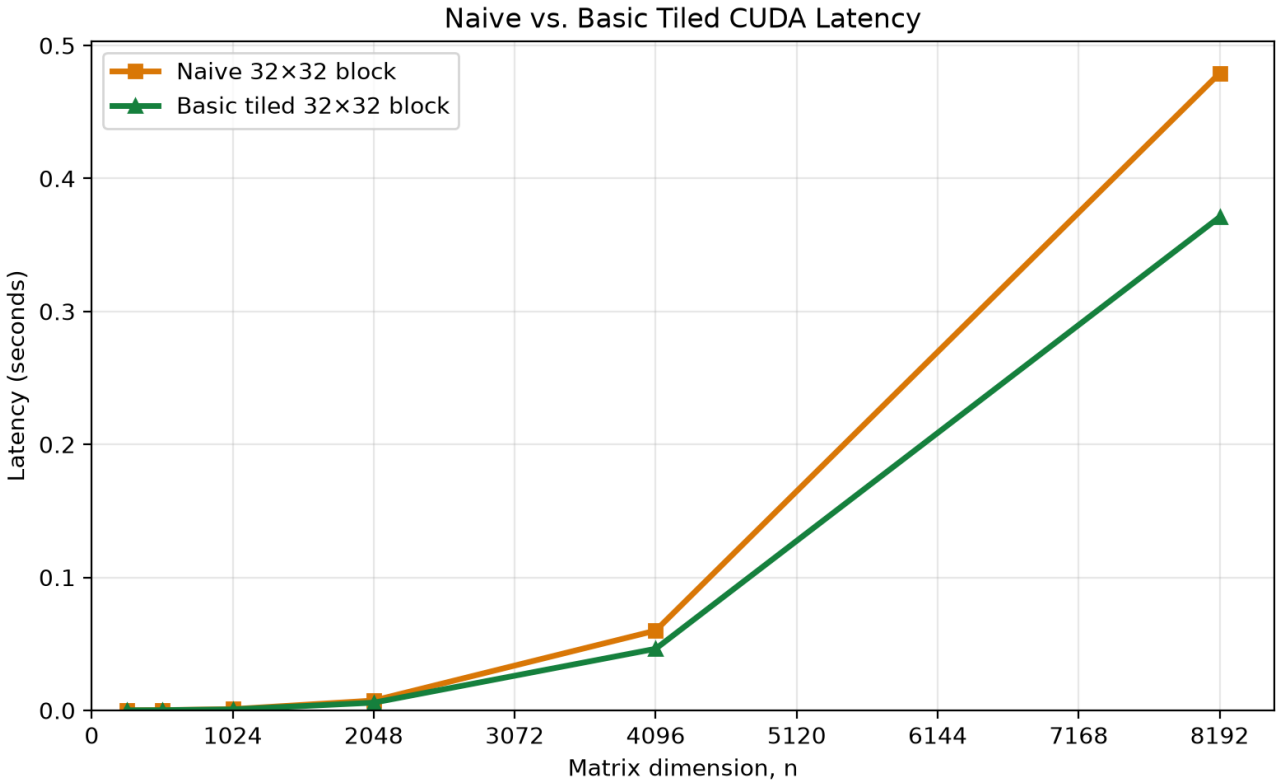
We have been measuring our kernels with **latency**, but is there some other metric that better describes "performance"?

How fast *can* we be?

Given our hardware constraints, how fast (theoretically) can we do our SGEMM operations?

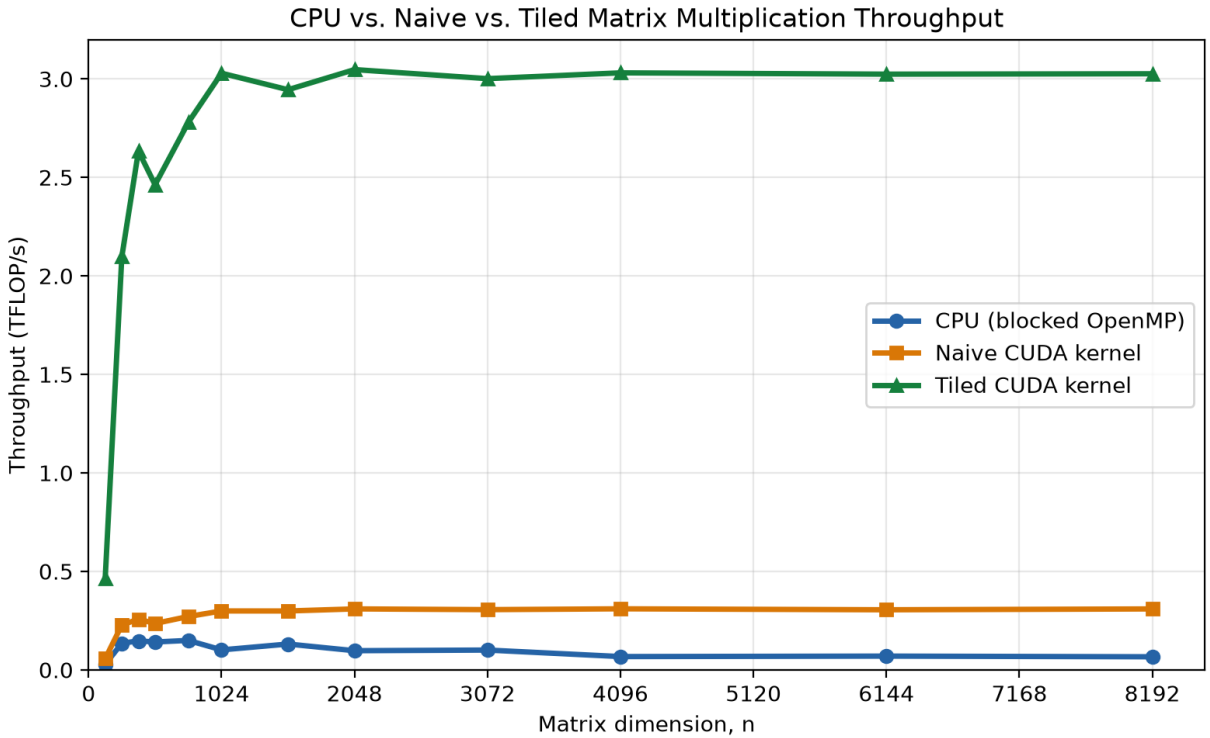
Why latency alone isn't enough

- Latency tells us how long one kernel launch takes, but it depends on the problem size.



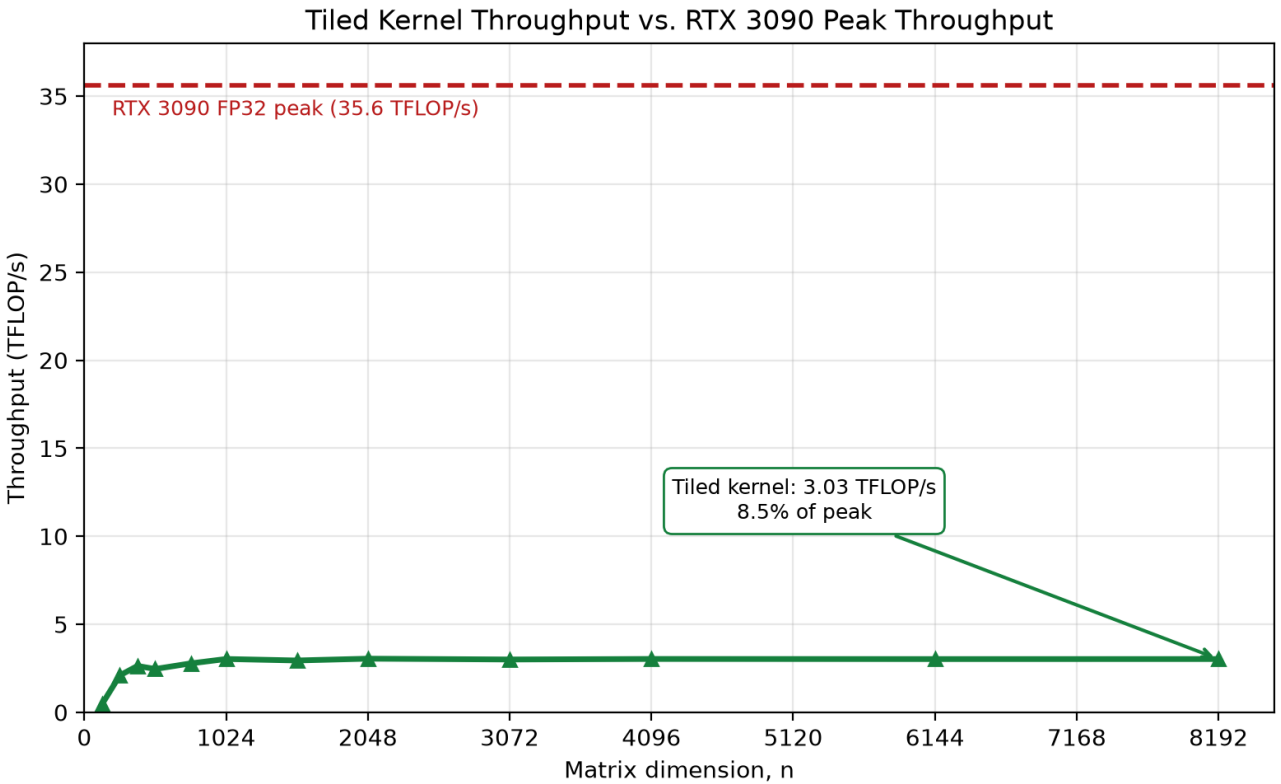
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Why throughput

Throughput tells us

1. A general performance metric, no matter the problem size
2. How well we are utilizing hardware

Throughput as a performance measure

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Example

Peak throughput of our RTX 3090 is 35.6 TFLOP/s. We observed 360ms or 0.36s of latency for $n = 8192$, where the total FLOPs required is 1.10 TFLOPs. Thus, we have an observed throughput of

$$\frac{1.10\text{TFLOPs}}{0.36\text{s}} \approx 3.05\,\text{TFLOP/s}.$$

Roughly 8.5% of peak throughput.

Characterizing our kernels data use

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1. Throughput characterizes how well our kernel utilizes the **compute hardware**.
2. One factor that impacts final throughput is how efficiently our kernels use data.

How can we quantify how "well" our kernel uses data?

We observed that we can improve the performance of our SGEMM kernel with tiling, an optimization that improves *data use*.

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Arithmetic Intensity: Characterizing our kernels

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We define the **Arithmetic Intensity** (AI) of our kernel as the number of FLOPs done per bytes moved from DRAM.

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Example (Naive kernel)

For $n, m, k = 8192$, in the worst case, the naive kernel loads (from DRAM)

$$n \times m \times 2k \times 4 \text{ bytes} \approx 4398 \text{ GB}.$$

We do 1.10 TFLOPs of work, so we have an AI of

$$\text{AI} = \frac{1.10 \text{ TFLOP}}{4.398 \text{ TB}} \approx 0.25 \text{ FLOP/byte}.$$

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Example (Tiled kernel)

For $n, k, m = 8192$, in the worst case, our tiled kernel loads (from DRAM)

$$\frac{m}{32} \times \frac{n}{32} \times 2\left(\frac{k}{32}\right) \times 32 \times 32 \times 4 \text{ bytes} = 137.4 \text{ GB}$$

With a total work count of 1.10 TFLOP, its arithmetic intensity is

$$\text{AI} = \frac{1.10 \text{ TFLOP}}{137.7 \text{ GB}} \approx 8.0 \text{ FLOP/byte}.$$

The Memory Roof

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Our tiling kernel has AI of 8.0 FLOP/byte, meaning it does 8 floating point operations for every byte of data moved from DRAM.
At $935.8\,\mathrm{GB/s}$ of bandwidth, our **memory roof** is

$$935.8\,\mathrm{GB/s} \times 8.0\ \mathrm{FLOP/byte} \approx 7.49\,\mathrm{TFLOP/s}.$$

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This is much lower than the peak throughput of $35.6\,\mathrm{TFLOP/s}$. **No matter how we optimize this kernel, without increasing AI, we will never achieve greater than** $7.49\,\mathrm{TFLOP/s}$.

Hypothetical Kernel

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$$935.8\,\mathrm{GB/s} \times 16.0\ \mathrm{FLOP/byte} \approx 15.0\,\mathrm{TFLOP/s}.$$

Since $15.0\,\mathrm{TFLOP/s} < 35.6\,\mathrm{TFLOP/s}$, this kernel will still be memory bound.

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AI (FLOP/byte)	Upper bound throughput (TFLOP/s)
8	7.49
16	14.97
32	29.95
36	33.69
38	35.56
40	37.43

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38	35.56
40	37.43

Wait! Our peak throughput is 35.6 $\mathrm{TFLOP/s}$! At AI ≥ 38, we are **compute bound**. Our data can finally keep up with the CUDA Cores!

Ridge Point

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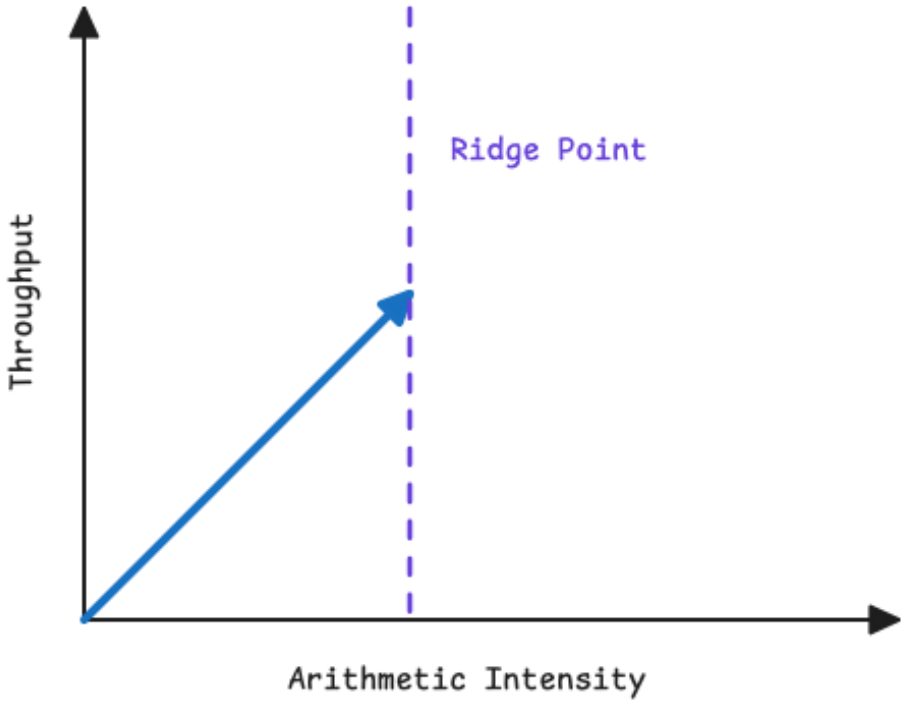
AI $= 38$ FLOP/byte is known as the **ridge point** *RP*, and can be found by solving:

$$\begin{aligned}
\text{Data roof} &= \text{Peak throughput} \\
\text{DRAM Bandwidth} \times \text{RP} &= \text{Peak throughput} \\
\text{RP} &= \frac{\text{Peak throughput}}{\text{DRAM Bandwidth}} \\
\text{RP} &= \frac{35.6\,\mathrm{TFLOP/s}}{935.8\,\mathrm{GB/s}} \approx 38.0 \text{ FLOP/s}
\end{aligned}$$

Roofline Model

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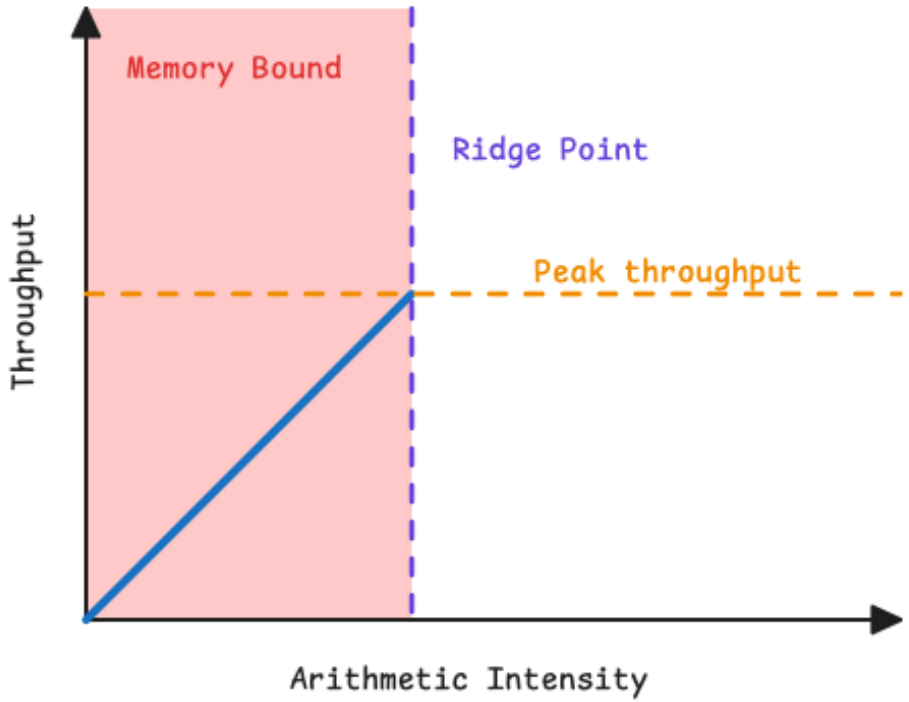
For a given system, we can predict a the **upper bound** of throughput of a kernel given its AI.



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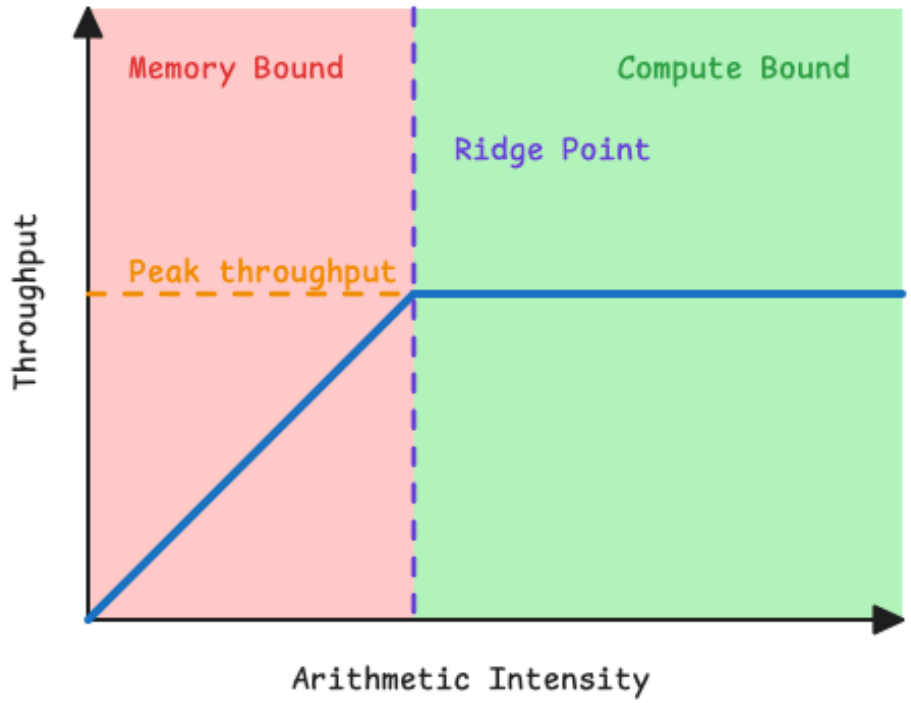
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For a given system, we can predict a the **upper bound** of throughput of a kernel given its AI. Up to the **ridge point**, we are **memory bound**, increasing AI will raise the upper bound of throughput. Past that, we are **compute bound**, increasing AI will not raise the upper bound of throughput.



Roofline Model of our RTX 3090

